

SokaSoko — Org Staff Delegation + GUARDIAN Caps (Internal QA)

Version 1.0 · Internal QA

What you're testing

The org-staff delegation model + tightened GUARDIAN caps. ACADEMY, CLUB, and SCHOOL orgs can invite GUARDIAN accounts as staff via a two-way handshake. Roles carry different delegated privileges: OWNER/MANAGER/COACH inherit the org tier, OTHER is chat-only, SPORTS_TEACHER is fixed at COACH GOLD (School only). Seat quotas come from FEATURE_CAPS[type][tier].staffSeats (0/3/5 for Academy and Club by tier, flat 3 for School). Standalone GUARDIANs are capped at 15 minors + 5 referee-minors and blocked from posting trials/tournaments and scheduling matches.

Prerequisites

- Backend commit 1edda28 or later on main.
- Multiple test accounts: ACADEMY (Standard, Gold, Platinum), CLUB (Gold), SCHOOL, plus several GUARDIAN accounts to invite.
- MongoDB access to seed / inspect OrgStaffLink and User.staffOverQuotaSince.
- Ability to trigger runStaffQuotaSweep manually for quota tests (or wait for the daily 08:30 cron).

Environment

- Backend: <https://sokasoko.onrender.com> — 1edda28+.
- Endpoints: POST/GET under /v1/users/:orgId/staff and /v1/staff-invitations.
- GET /v1/subscriptions/me returns a top-level `delegation` block.

Test cases

1. Invite endpoint — happy path

- As ACADEMY Gold, POST /v1/users/:orgId/staff/invite { guardianId, role: 'MANAGER' }.

Expected: 201 with new OrgStaffLink { status: 'PENDING' }. Notification created on the guardian with type=STAFF_INVITE and metadata.linkId.

2. Role restriction — SCHOOL can only invite SPORTS_TEACHER

- As SCHOOL, POST invite with role='MANAGER'.

Expected: 400 with 'SCHOOL can only invite SPORTS_TEACHER role'.

3. Role restriction — Academy cannot use SPORTS_TEACHER

- As ACADEMY, POST invite with role='SPORTS_TEACHER'.

Expected: 400 with 'SPORTS_TEACHER is a SCHOOL role only'.

4. Seat cap on Standard (0)

- As ACADEMY Standard, POST an invite.

Expected: 403 with reason=STAFF_SEATS_ZERO.

5. Seat cap on Gold (3)

- As ACADEMY Gold with 3 ACTIVE staff, POST a 4th invite.

Expected: 403 with reason=STAFF_SEATS_FULL, cap=3, current=3.

6. Seat cap counts PENDING toward quota

- As ACADEMY Gold with 2 ACTIVE + 1 PENDING, POST another invite.

Expected: 403 STAFF_SEATS_FULL (quota is ACTIVE + PENDING = 3, which equals cap).

7. One-to-one enforcement — API-level

- Guardian G is ACTIVE staff at Academy A. Academy B POSTs invite for G.

Expected: 409 with reason=STAFF_ALREADY_LINKED.

8. One-to-one enforcement — DB level

- Attempt to bypass the API check and directly create two ACTIVE OrgStaffLinks with the same staff id.

Expected: Second create fails with E11000 (partial unique index on { staff: 1 } filtered by status='ACTIVE').

9. Accept flow

- Guardian POSTs /v1/staff-invitations/:linkId/accept on their PENDING invite.

Expected: Link.status → ACTIVE. Guardian.acceptedAt stamped. If org has downgraded in the meantime and now cap is exceeded, accept returns 403 STAFF_SEATS_FULL.

10. Decline flow

- Guardian POSTs /v1/staff-invitations/:linkId/decline.

Expected: Link.status → DECLINED. Seat freed up.

11. Remove flow

- Org POSTs /v1/users/:orgId/staff/:linkId/remove on an ACTIVE staff.

Expected: Link.status → DISABLED, disabledAt stamped, disabledReason='Removed by org'. Notification created on the staff with type=STAFF_REMOVED.

12. GUARDIAN minor count cap

- GUARDIAN with 15 minors POSTs /v1/users to create a 16th PLAYER minor.

Expected: Error: 'Umefikia kikomo cha wachezaji 15.' User NOT created.

13. GUARDIAN referee count sub-cap

- GUARDIAN with 5 referee-minors POSTs /v1/users to create a 6th REFEREE minor.

Expected: Error: 'Umefikia kikomo cha waamuzi 5 kama mlezi.'

14. Bare guardian cannot post trial

- GUARDIAN with no ACTIVE staff link POSTs /v1/trials with organizer=self.

Expected: 403 reason=GUARDIAN_TRIAL_CREATION_BLOCKED.

15. Bare guardian cannot post tournament

- GUARDIAN POSTs /v1/tournaments with organizer=self.

Expected: 403 reason=GUARDIAN_TOURNAMENT_CREATION_BLOCKED.

16. Bare guardian cannot schedule match

- GUARDIAN POSTs /v1/matches with scheduledBy=self.

Expected: 403 reason=GUARDIAN_MATCH_SCHEDULE_BLOCKED.

17. Sports Teacher delegation

- GUARDIAN is ACTIVE SPORTS_TEACHER at a SCHOOL.
- POST /v1/trials with organizer=.
- POST /v1/matches with scheduledBy=.

Expected: Both accept — getEffectiveContext returns userType=COACH tier=GOLD caps=COACH.GOLD. Guardian effectively behaves as a paid COACH while linked.

18. Academy Coach delegation inherits Platinum

- Academy on Platinum invites a guardian as role=COACH; guardian accepts.
- Guardian POSTs a trial or tournament with maxTeams=20 as organizer=self.

Expected: Accepted — delegated context resolves to userType=ACADEMY tier=PLATINUM caps=ACADEMY.PLATINUM, which has maxTournamentTeams=null (unlimited).

19. Academy OTHER role has no elevation

- Guardian is ACTIVE staff with role=OTHER at an Academy.
- Guardian POSTs a trial.

Expected: 403 GUARDIAN_TRIAL_CREATION_BLOCKED — OTHER role doesn't inherit trial-post powers (caps=null in the delegation resolution; guardian falls back to bare).

20. Delegation surfaced on /me

- GET /v1/subscriptions/me?userId=&userType;=GUARDIAN.

Expected: Response includes `delegation: { org, orgType:'SCHOOL', orgTier:'GOLD', role:'SPORTS\_TEACHER', effectiveUserType:'COACH', caps }`. For a bare guardian: delegation:null.

21. Over-quota grace sweep

- Academy on Platinum with 5 ACTIVE staff downgrades to Gold.
- Manually trigger runStaffQuotaSweep (or wait 24h).

Expected: First run: User.staffOverQuotaSince stamped. Owner notification 'Umezidi kikomo cha wafanyakazi'. Subsequent runs within 5 days: no action. After 5+ days over quota: all 5 links flip to DISABLED, each staff gets a STAFF_DISABLED notification, staffOverQuotaSince unset.

22. Recovery from over-quota

- Academy over quota, owner removes 2 staff proactively.
- Next scheduler run.

Expected: activeCount ≤ cap → User.staffOverQuotaSince unset. No further notifications. Staff remain ACTIVE.

23. Non-blocking enforcement failures

- Simulate mongoose model resolution failure (e.g., OrgStaffLink not registered).
- Hit any delegation-aware endpoint.

Expected: Delegation helpers catch and return null. Endpoint proceeds with bare-user context — does NOT crash. Verify by logs.

How to report a bug

Field	Example
Test case #	9 — Accept flow
Device / OS	Pixel 6a · Android 14

App version	1.0.1+14
Backend commit	1edda28
Account	ACADEMY · TFH-A-A000005
Endpoint	POST /v1/staff-invitations/:linkId/accept
Expected	200 with link.status=ACTIVE
Actual	403 STAFF_SEATS_FULL despite empty seats
Screenshot / repro	attached
Render log excerpt	attached

Known limitations (not bugs)

- No mobile UI yet — owners and guardians run through API endpoints. Building the manage-staff screen + invite-inbox is the follow-up mobile pass.
- Report request and media POST delegation is NOT yet wired. A guardian who is Academy Coach staff will still be blocked from report generation because those endpoints haven't been updated to consult `getEffectiveContext`. Trial/tournament/match are wired.
- Guardian chat-scope restriction (limit chat to their minors' network) is not enforced yet.
- `/v1/users/eligible-scouts` continues to pull from `sports_teacher_1/2` legacy fields only. Doesn't yet include `SPORTS_TEACHER OrgStaffLinks` — a follow-up.
- Staff quota sweep runs at 08:30 daily. To test the 5-day grace, either time-travel the `User.staffOverQuotaSince` field manually in DB or invoke `runStaffQuotaSweep` directly from a node REPL.
- Downgrading during a PENDING invite → the pending invite becomes counted against the new (smaller) cap on Accept; the accept can 403. Expected behaviour; the guardian sees 'seats full, contact org'.